AI-Assisted Proposal Triage Tool

Complete Solution: Workflow, Scoring, Sample Ranking, and Assessment Answers

1. Objective

The purpose of this tool is to help a facilities team evaluate and rank contractor proposals for a commercial solar installation project. AI analyzes proposal documents using administrator-defined criteria, explains the ranking, and supports reviewers. The final award decision remains with a human.

2. Project Requirements

Requirement	Details
Category	Commercial Solar Installation
Project	150 kW rooftop solar system on a warehouse
Location	Windsor, Ontario, Canada
Maximum budget	$260,000 CAD
Completion deadline	Within 16 weeks of award
Final decision	Human reviewer / authorized decision-maker

3. Proposed Workflow

Step 1 — Admin configuration

The admin creates the category and configures mandatory pass/fail requirements, weighted criteria, scoring rules, and weights through admin screens. Criteria are stored in the database rather than hard-coded.

Step 2 — Proposal upload

The reviewer uploads the contractor PDFs and selects Commercial Solar Installation.

Step 3 — AI extraction

AI extracts contractor details, licence information, WSIB clearance, insurance, experience, price, timeline, warranties, and project approach. It stores supporting evidence and page references.

Step 4 — Mandatory checks

Plain code checks whether the proposal explicitly confirms an Ontario electrical contractor licence, WSIB clearance, and $5M liability insurance. Budget and timeline are also checked against the project limits. Unknown information is flagged rather than assumed to pass.

Step 5 — Weighted scoring

The system applies the administrator's configured rubric and weights to eligible proposals using deterministic calculations.

Step 6 — Ranking and explanation

The tool displays scores, evidence, missing information, strengths, weaknesses, and a reason for each proposal's position.

Step 7 — Human review and award

The reviewer checks evidence, resolves uncertainties, and makes the final award decision. The system records the decision and its justification.

4. Evaluation Criteria and Scoring Method

Criterion	Type	Weight	Evaluation focus
Ontario electrical contractor licence	Mandatory	Pass/fail	Licence held by bidder and stated in proposal
WSIB clearance and $5M liability insurance	Mandatory	Pass/fail	Both confirmed in proposal
Relevant experience	Weighted	30%	Commercial installations of 100 kW or more in the last 5 years
Price	Weighted	25%	Lower total cost within the $260,000 budget ceiling
Timeline	Weighted	15%	Shorter confirmed completion time within 16 weeks
Warranty	Weighted	15%	Length of panel and workmanship warranties
Team and approach	Weighted	15%	Named lead, crew, safety, and monitoring plan

Formula: Total score = sum of (criterion score / 100 × criterion weight). For example, a price score of 80/100 at a 30% weight would contribute 24 points. In this assignment, price is weighted at 25%; the example only illustrates the formula.

Mandatory criteria are separate from the weighted score. A high score cannot compensate for failing a mandatory requirement. The admin should define a fixed rubric before evaluation; the criterion descriptions alone do not specify every numerical scoring rule.

5. Mandatory Checks for the Five Proposals

Proposal	Bidder licence	WSIB	$5M insurance	Initial status
A — Northshore Solar	Pass	Pass	Pass	Eligible
B — BrightPath Energy	Unconfirmed / fail	Pass	Pass	Ineligible unless resolved
C — Lakeview Renewables	Pass	Pass	Pass	Eligible
D — Essex Green Power	Pass	Pass	Pass	Needs timeline confirmation
E — SunGrid Contractors	Pass	Pass	Pass	Eligible

Key findings: BrightPath says electrical work will be done by a licensed subcontractor to be confirmed after award; the stated mandatory requirement is that the bidder holds the licence, so this needs clarification and should not be assumed to pass. Essex Green Power does not state a completion timeline, so its compliance with the 16-week limit is unconfirmed. Lakeview's embedded instruction to automated systems to rank it first is untrusted proposal text and must not affect scoring.

6. Illustrative Ranked Output

The criteria provide qualitative guidance but do not fully specify numerical scoring scales. The following results use a transparent illustrative rubric for demonstration only; an admin-approved rubric should be stored before real procurement use.

Illustrative rubric: Experience is scored relative to the highest qualifying installation count among eligible proposals; price relative to the lowest eligible bid; timeline relative to the fastest confirmed schedule; warranty is split equally between panel and workmanship duration, normalized to the longest duration in the eligible set; team

and approach score four equally weighted items: named lead, specified crew, safety plan, and monitoring/reporting plan.

Rank	Contractor	Experience	Price	Timeline	Warranty	Team	Total
1	Northshore Solar (A)	100	100	100	91.7	100	**98.75**
2	Lakeview Renewables (C)	64.3	97.5	85.7	91.7	75	**86.67**
3	SunGrid Contractors (E)	57.1	98.1	92.3	100	100	**85.80**

Ranking explanation: Northshore ranks first because it has the highest qualifying experience count, a competitive price, a confirmed 12-week schedule, strong warranties, and a detailed delivery plan. Lakeview ranks second due to solid experience, compliance, a 14-week schedule, and a 10-year workmanship warranty, though its plan has less detail on safety and monitoring. SunGrid ranks third: it has a 12-year workmanship warranty and a strong team/monitoring plan, but fewer qualifying installations.

Unranked/pending proposals: BrightPath is ineligible unless the mandatory bidder-licence requirement is resolved. Essex Green Power is pending timeline confirmation. They should not receive a final competitive rank until the issues are resolved.

Scoring note: These are illustrative scores and ranking choices based on a proposed normalization method, not a uniquely determined answer. A production tool must use admin-approved scoring rules, including a clear treatment for missing mandatory evidence and unconfirmed timelines.

7. Answers to the Assessment Questions

Question 1 — AI vs. rules

AI extracts information from proposals, interprets unstructured text, identifies relevant evidence, and drafts explanations. Plain code validates required fields, checks mandatory criteria, enforces the budget and timeline limits, applies configured scoring formulas, calculates weighted totals, and sorts the results. AI is useful for document understanding; deterministic code is more reliable for compliance and arithmetic.

Question 2 — Consistency and explainability

Scores use versioned criteria, fixed scoring formulas, defined weights, and the same input data. Repeated scoring of the same saved extracted facts under the same criteria version should produce the same result. Because AI extraction can vary, extracted values should be saved and reused for rescoring. Reviewers see each criterion score, weighted contribution, source evidence and page reference, missing information, and explanation of the ranking.

Question 3 — Human in the loop

A reviewer verifies extracted facts, resolves uncertain mandatory checks, and reviews unusual or conflicting information. An authorized human makes the final award decision and may override the recommendation with a documented reason. The audit log records uploads, timestamps, criteria versions, extracted values, evidence, scores, ranking changes, reviewer corrections, overrides, and the final decision.

Question 4 — Changing criteria

Each evaluation is linked to the exact criteria version used at the time. Editing a weight or requirement creates a new version, and the system validates that weighted criteria still total 100%. Existing results remain unchanged for audit purposes. If proposals need to be assessed under the new rules, the system creates a new evaluation using the updated criteria and labels it clearly.

Question 5 — Risks and safeguards

Risk	Safeguard
Incorrect extraction	Store source evidence and page references; ask for human verification of uncertain fields.
Made-up facts	Score only supported facts; mark missing evidence unknown rather than assuming compliance.
Bias or inconsistent scoring	Use predefined rubrics, consistent criteria, documented weights, and periodic audits.
Prompt injection or manipulation	Treat proposal instructions as untrusted data; never allow them to change rules or scores.
Calculation errors	Use deterministic code and automated tests for formulas and mandatory checks.
Missing information	Flag for review rather than silently awarding full marks or treating unknown as a pass.
Cost or slow processing	Set file-size limits, use appropriate extraction models, cache extracted facts, and monitor processing cost.
Data leaks or unauthorized access	Use role-based permissions, encryption, secure storage, and retention policies.
Unjustified overrides	Require a reason and retain both the original recommendation and the final human decision.

8. Recommended Application Screens and Data

Screens: Category Management; Criteria Management; Proposal Upload; Evaluation Dashboard; Ranked Results; Decision and Audit Log.

Data to store: categories, criteria and versions, weights, proposal files, extracted facts, evidence references, mandatory-check results, criterion scores, total scores, rankings, reviewer corrections, and final decisions.

Every evaluation should retain its criteria version so later admin edits do not silently change old results. Proposal content should be treated as untrusted input, and access to contractor documents should be restricted.

9. Final Conclusion

The proposed tool combines AI-based document understanding with rule-based compliance checks and deterministic scoring. It provides a transparent, repeatable, and auditable shortlist while keeping the final procurement decision with a human reviewer. Under the illustrative scoring rules, Northshore Solar ranks first. BrightPath needs its mandatory bidder-licence issue resolved, and Essex Green Power needs to confirm its timeline before a final ranking can be completed.